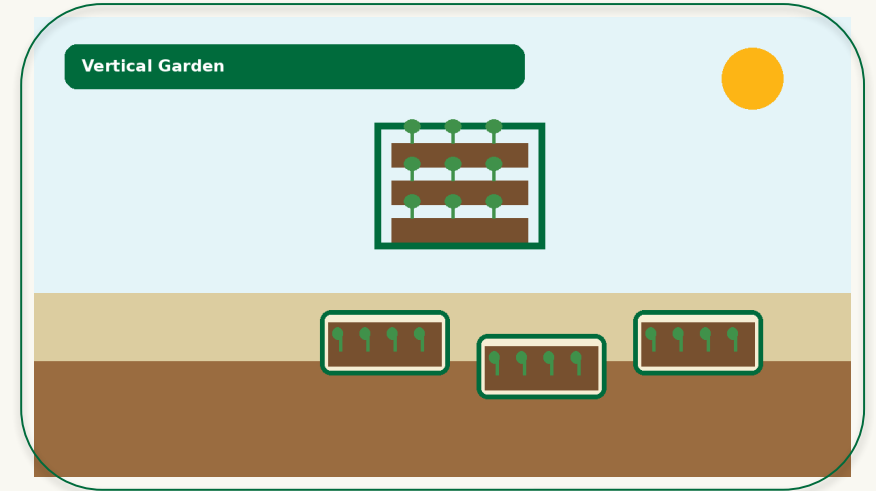


Vertical Garden

Growing Upward

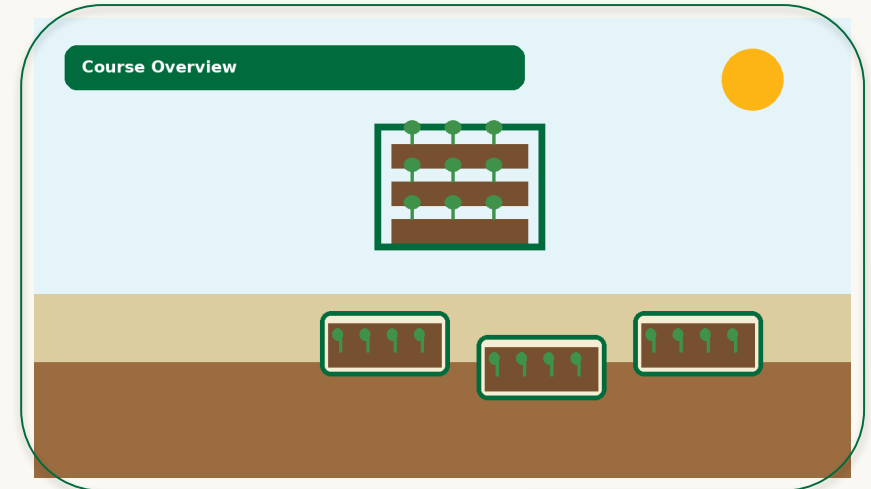
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Learning progression: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system and its educational purpose; teach all ten technical topics at Primary 4-6 level; guide safe practical work and measurement; connect science, mathematics and agriculture; maintain useful records; help pupils interpret observations without overclaiming.
- Core materials: teacher guide, system diagram, child-safe tools, labels, clipboards, rulers/tapes, graduated containers or measuring devices where relevant, observation/data sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson, measurement or investigation plan.



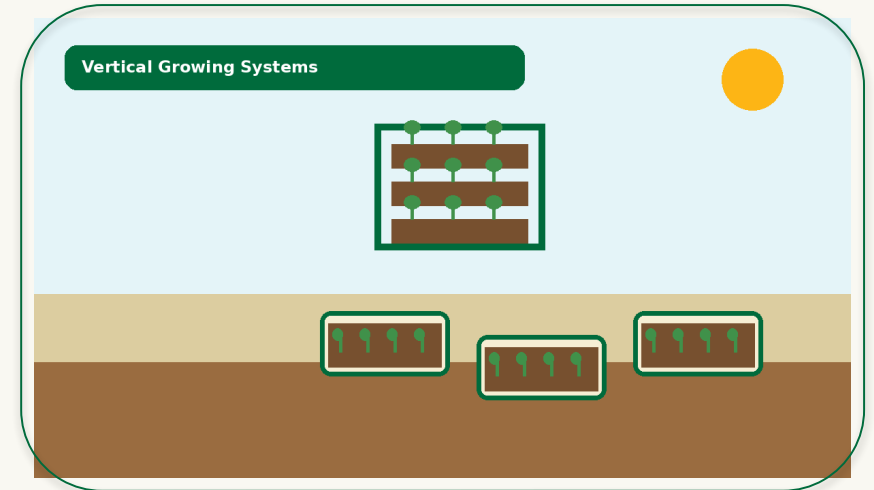
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Vertical Growing Systems

Teacher knowledge and practical application

- Define vertical growing as arranging plants in stacked, wall-mounted, tower or tiered systems to use upright space.
- Identify structural frame, containers, growing media, irrigation and drainage.
- Discuss advantages and constraints such as access, loading and uneven drying.
- Inspect stability before every pupil activity.
- Keep all pupil tasks within safe reach from ground level.



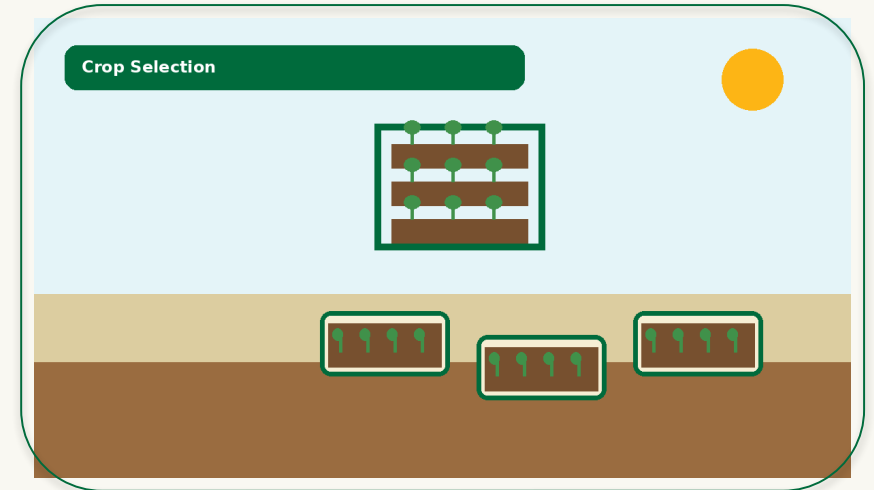
TEACHER PRACTICE

- Evaluate one vertical system for structure, access and crop space.

Crop Selection

Teacher knowledge and practical application

- Choose compact crops suited to container volume, root depth, sunlight and harvest method.
- Consider mature plant weight and avoid crops that can destabilize the structure.
- Use locally appropriate leafy vegetables, herbs and other approved crops.
- Plan crop placement before planting.
- Avoid thorny or highly irritating plants in pupil-access zones.



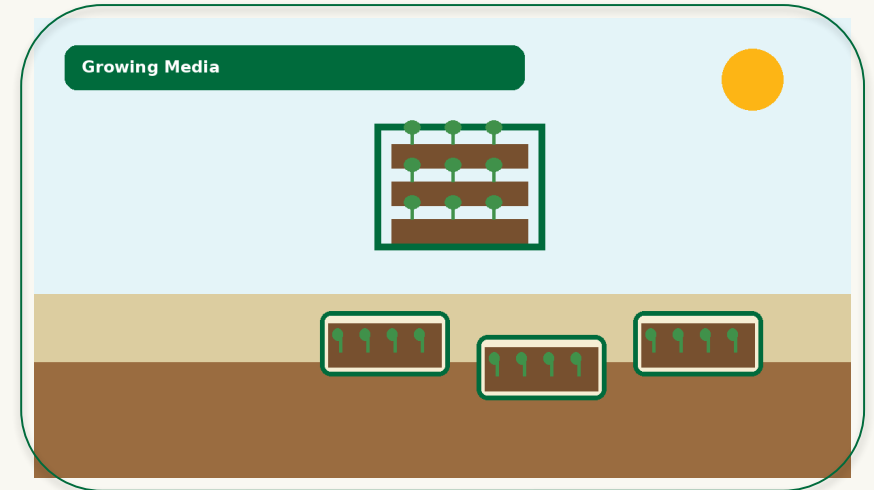
TEACHER PRACTICE

- Prepare an upper-middle-lower crop placement plan.

Growing Media

Teacher knowledge and practical application

- Use clean, lightweight media with suitable water-holding and drainage characteristics.
- Explain why vertical containers may have smaller root volumes than ground beds.
- Avoid excessive compaction.
- Monitor settling and exposed roots.
- Use only mature, approved amendments.



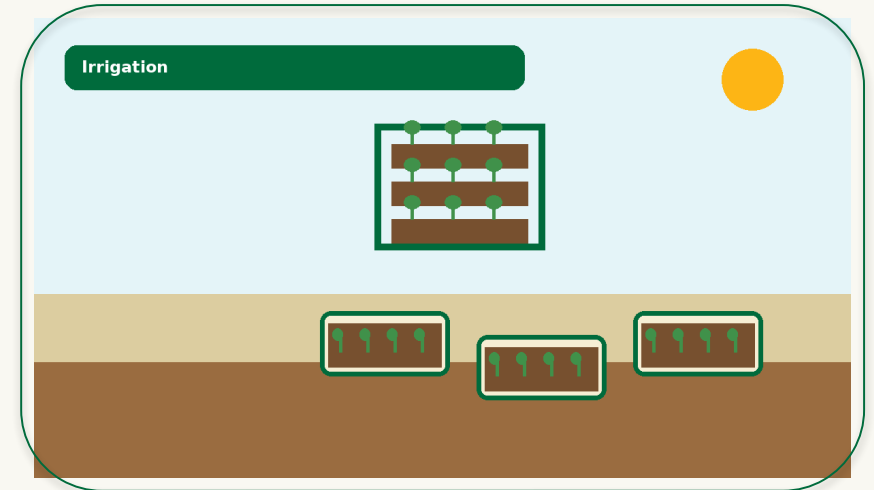
TEACHER PRACTICE

- Prepare a simple media observation activity.

Irrigation

Teacher knowledge and practical application

- Recognize that containers at different levels may receive or retain different amounts of water.
- Use controlled drip or careful hand watering.
- Check moisture at upper, middle and lower levels.
- Avoid excessive runoff from one level into another.
- Record recurring dry or wet zones.



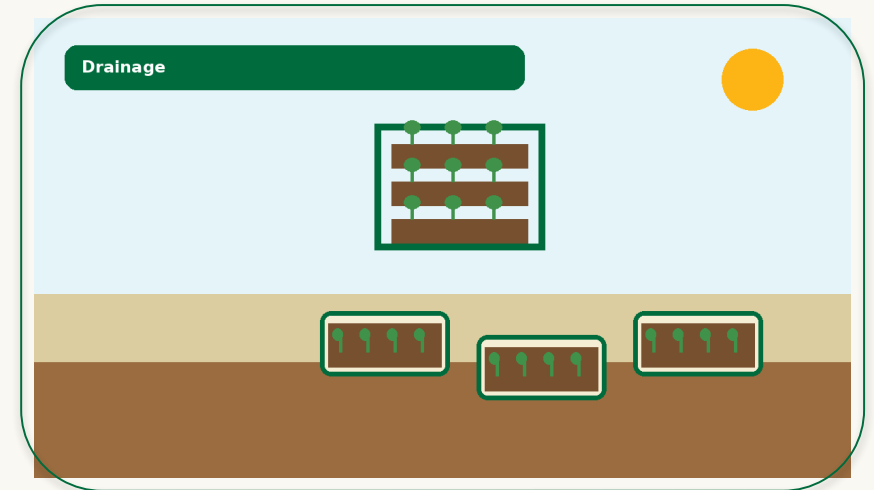
TEACHER PRACTICE

- Develop a three-level moisture-check routine.

Drainage

Teacher knowledge and practical application

- Explain why excess water must leave containers safely.
- Inspect drainage openings and collection channels.
- Prevent stagnant water and slippery paths.
- Observe whether lower levels receive unwanted runoff.
- Do not modify structural drainage during pupil sessions.



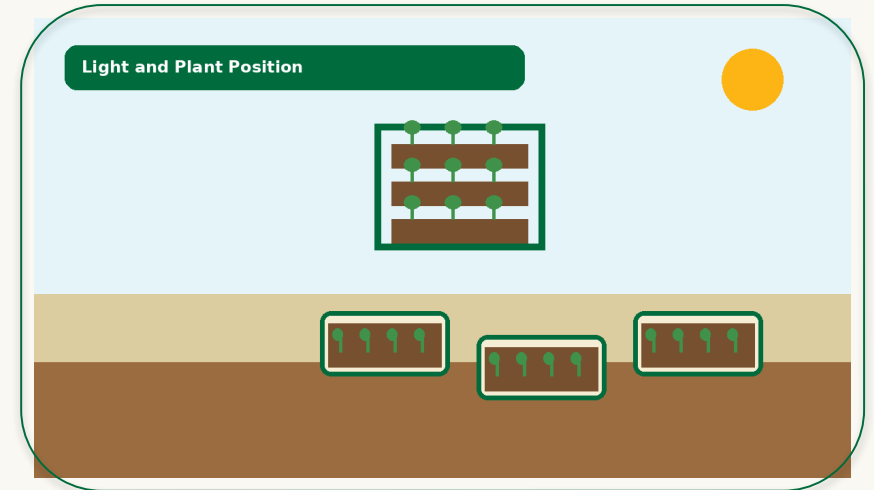
TEACHER PRACTICE

- Trace the drainage path after a controlled watering event.

Light and Plant Position

Teacher knowledge and practical application

- Observe how orientation and nearby structures affect light received by different levels.
- Compare visible growth without assuming light is the only cause of differences.
- Rotate movable units only if the system is designed for it and adult-managed.
- Choose crops suited to available light.
- Record sun/shade observations at selected times.



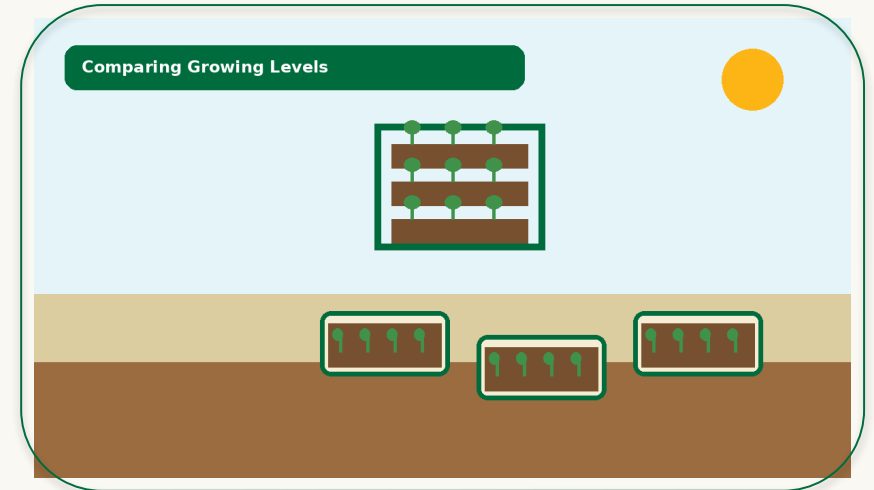
TEACHER PRACTICE

- Prepare a simple position-and-light observation chart.

Comparing Growing Levels

Teacher knowledge and practical application

- Measure the same crop at upper, middle and lower positions using consistent methods.
- Track height, leaf number, moisture or harvest as appropriate.
- Keep crop age and management similar when comparison is the goal.
- Record position with every measurement.
- Treat results as evidence from the school system, not a universal rule.



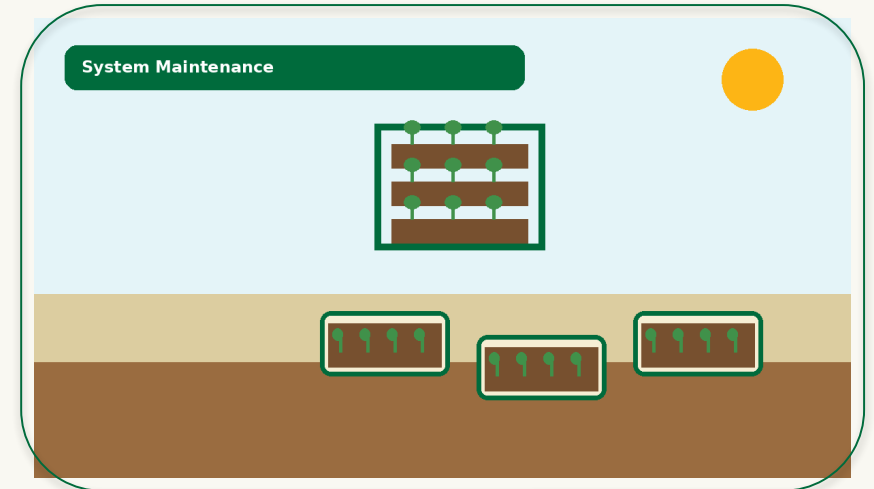
TEACHER PRACTICE

- Set up a three-level crop comparison.

System Maintenance

Teacher knowledge and practical application

- Inspect frames, fasteners, containers, irrigation and drainage weekly.
- Replace cracked containers and repair loose supports before pupil access.
- Remove dead material and confirmed weeds.
- Keep labels readable.
- Record structural and irrigation problems for adult action.



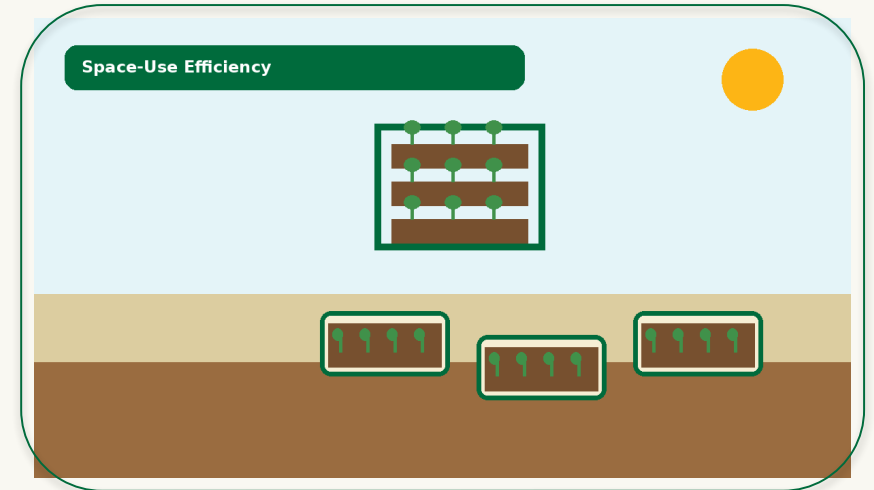
TEACHER PRACTICE

- Create a vertical-garden maintenance checklist.

Space-Use Efficiency

Teacher knowledge and practical application

- Compare growing container area or plant count with the ground footprint occupied by the system.
- Explain that efficient space use must still allow safe access, light and maintenance.
- Use simple ratios or counts appropriate to Primary 4-6.
- Compare with a ground bed without assuming vertical is always superior.
- Include pathway and maintenance requirements in discussion.



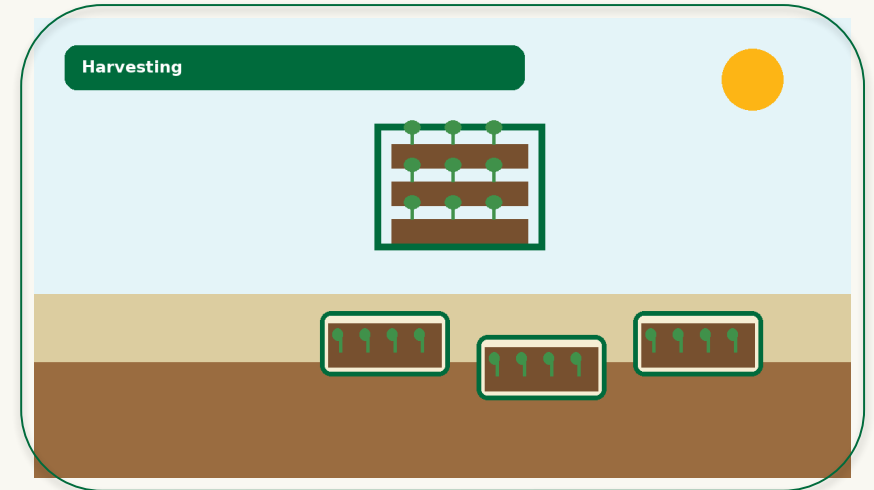
TEACHER PRACTICE

- Calculate a simple plants-per-footprint comparison.

Harvesting

Teacher knowledge and practical application

- Harvest from safe ground-level positions only.
- Use crop-specific readiness indicators.
- Do not allow climbing or pulling on the structure.
- Use clean containers and approved tools.
- Record harvest by level when this supports the learning objective.



TEACHER PRACTICE

- Completion task: plan a vertical-garden measurement and harvest lesson.